CIE AS PHYSICS (9702) • QUESTION PAPER

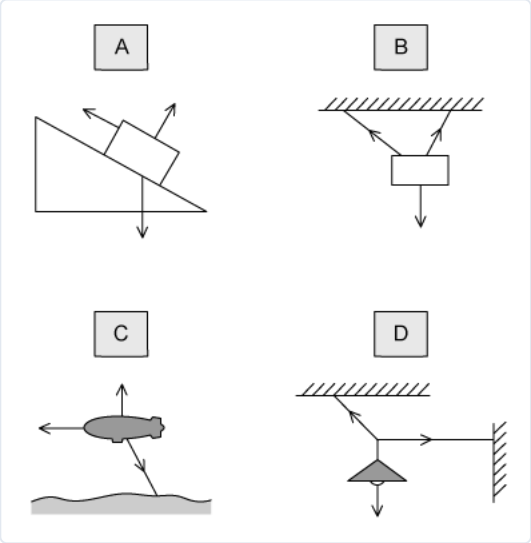
4.2 Equilibrium of Forces

Syllabus Section: 4. Forces, Density & Pressure **Paper Type:** Multiple Choice (Paper 1) **Total:** 23 Questions (23 Marks)

Question 1 Easy [1 mark]

The diagrams show the forces acting on different bodies.

Which body **cannot** be in equilibrium?



A

B

C

D

Question 2 Easy [1 mark]

A force of 5 N pushes a ball due north, and another force of 12 N pushes it due east.

What is the magnitude of the net force acting on the ball?

A 7 N

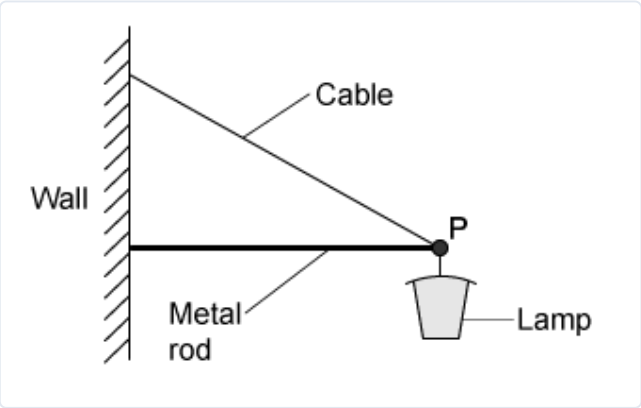
B 11 N

C 13 N

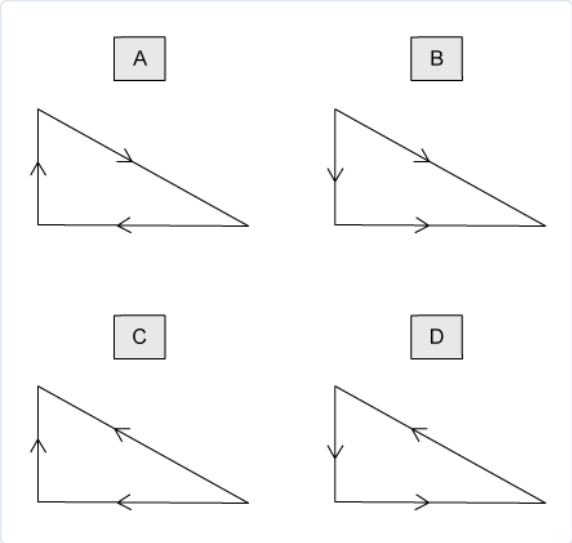
D 17 N

Question 3 Easy [1 mark]

A street lamp is fixed to a wall by a metal rod and a cable.



Which vector triangle represents the forces acting at point P?



A

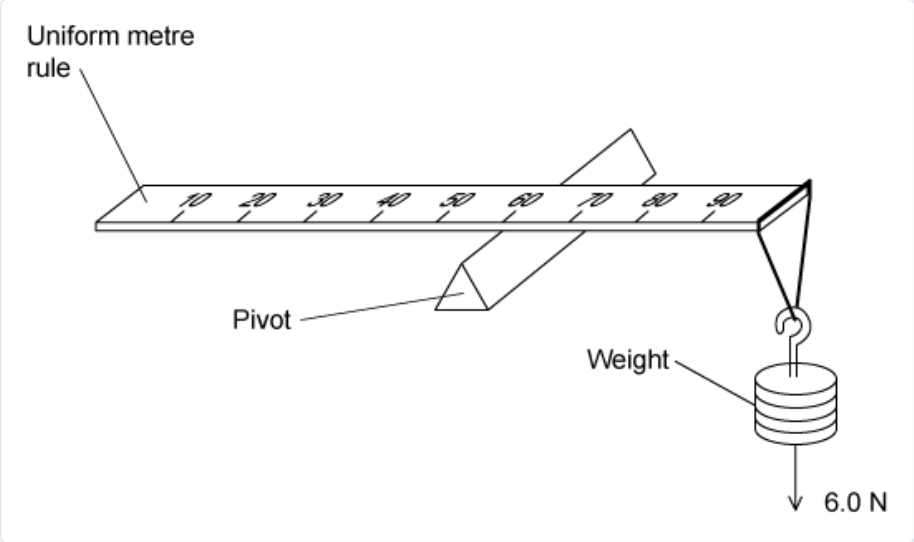
B

C

D

Question 4 Easy [1 mark]

A uniform metre rule of weight 4.0 N is pivoted at the 60 cm mark. A 6.0 N weight is suspended from one end, causing the rule to rotate about the pivot.



At the instant when the rule is horizontal, what is the resultant turning moment about the pivot?

	Resultant moment	Direction
A	2.0 N m	clockwise
B	2.0 N m	anticlockwise
C	2.8 N m	clockwise
D	2.8 N m	anticlockwise

A

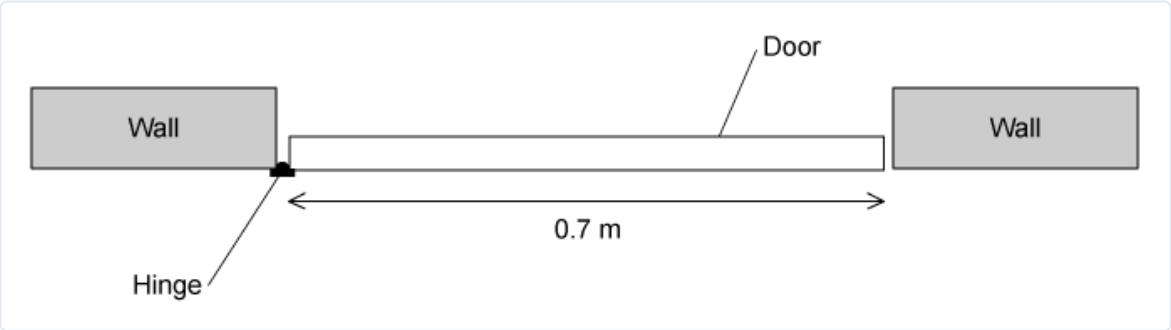
B

C

D

Question 5 Easy [1 mark]

The diagram shows a plan view of a door which requires a moment of 14 N m to open it.



What is the minimum force that must be applied at the door's midpoint to ensure it opens?

A 4.9 N

B 9.8 N

C 20 N

D 40 N

Question 6 Easy [1 mark]

Forces of 3 N, 4 N and 5 N act at one point on an object. The angles at which the forces act can vary.

What is the value of the minimum resultant force of these forces?

A 0 N

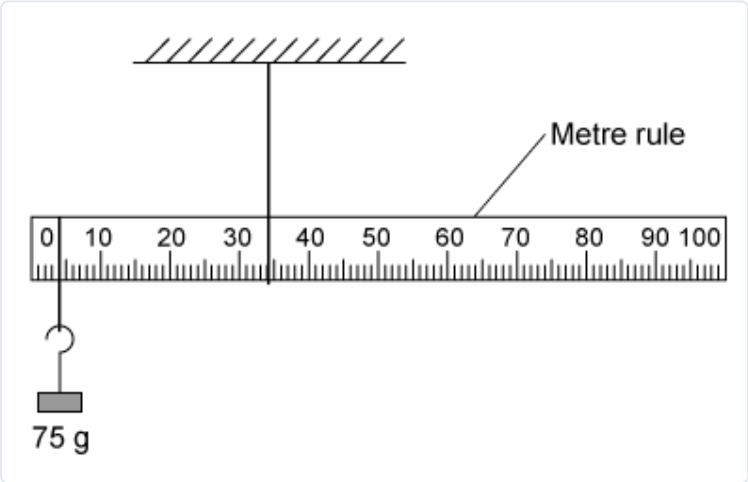
B between 0 and 2 N

C 2 N

D between 2 N and 4 N

Question 7 Medium [1 mark]

A uniform metre rule is hung from the ceiling as shown



The rule balances when a 75 g mass is hung from one end of the ruler

What is the mass of the metre rule?

A 38 g

B 51 g

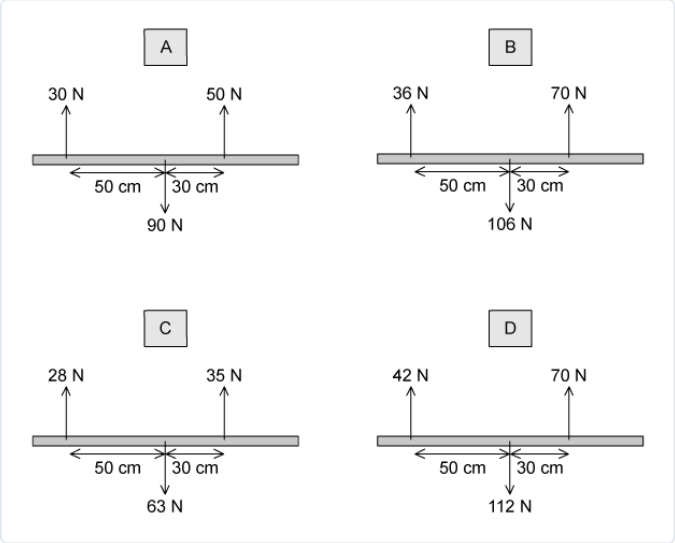
C 141 g

D 159 g

Question 8 Medium [1 mark]

Four beams of the same length each have three forces acting on them.

Which beam has both zero resultant force and zero resultant torque acting?



A

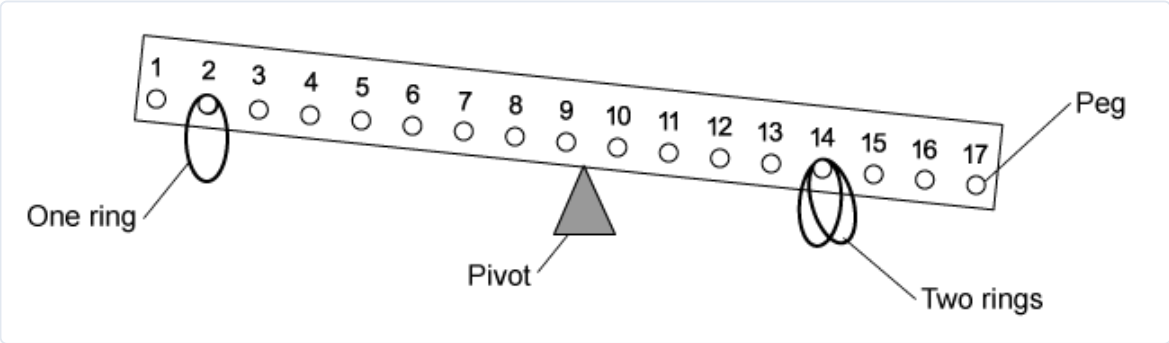
B

C

D

Question 9 Medium [1 mark]

The diagram shows a child's balancing game.



The wooden rod is uniform and all the rings are of equal mass. Two rings are hung on peg 14 and one on peg 2. The rod is pivoted at peg 9.

On which hook must a fourth ring be hung in order to balance the rod?

A 2

B 3

C 5

D 6

Question 10 Medium [1 mark]

An object, made from two equal masses joined by a light rod, falls with uniform speed through air. The rod remains horizontal.

Which statement about this system is correct?

A it is not in equilibrium because it is falling steadily

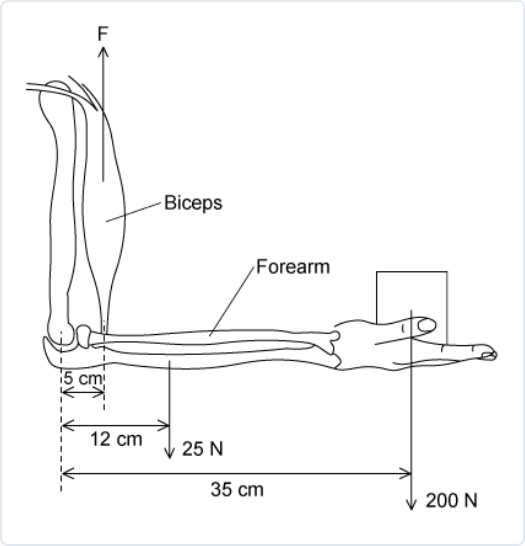
B it is not in equilibrium because it is in motion

C it is not in equilibrium because there is a resultant torque

D it is in equilibrium because there is no resultant force and no resultant torque

Question 11 Medium [1 mark]

A man holds a 200 N load stationary in his hand. The combined weight of the forearm and hand is 25 N. The forearm is held horizontal, as shown



What is the vertical force *F* needed in the biceps?

A 1340 N

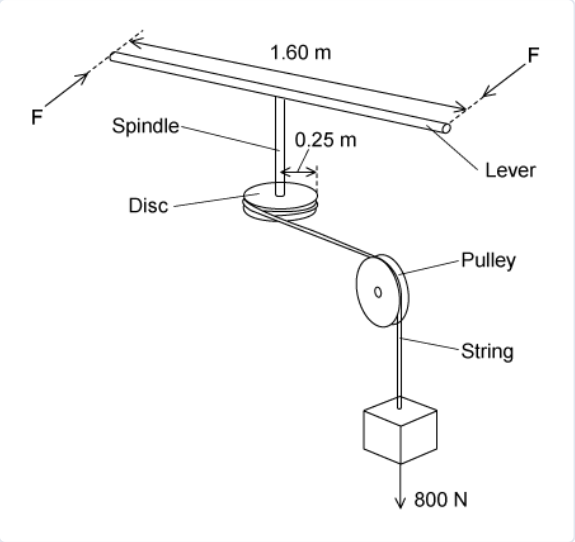
B 1400 N

C 1460 N

D 1520 N

Question 12 Medium [1 mark]

A spindle is attached at one end to the centre of a lever of length 1.60 m and at its other end to the centre of a disc of radius 0.25 m. A string is wrapped round the disc, passes over a pulley and is attached to an 800 N weight.



What is the minimum force *F*, applied to each end of the lever, that could lift the weight?

A 62.5 N

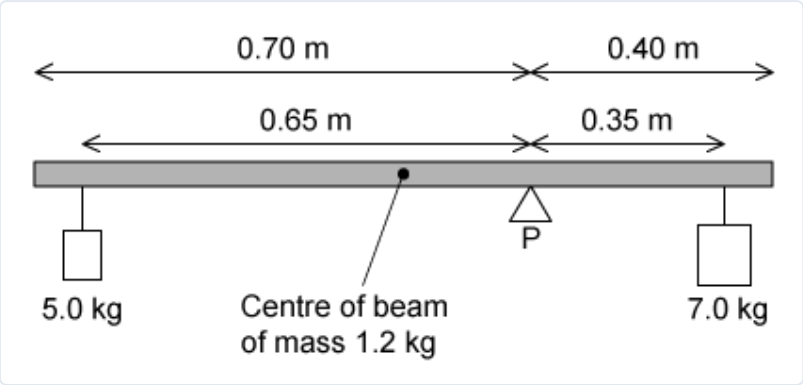
B 125 N

C 250 N

D 750 N

Question 13 Medium [1 mark]

A uniform beam of length 1.10 m and mass 1.2 kg is pivoted at P as shown. The pivot P is 0.40 m from one end. Loads of masses 5.0 kg and 7.0 kg are suspended 0.65 m and 0.35 m from the pivot, respectively, as shown.



What torque must be applied to the beam in order to maintain it in equilibrium?

A 0.62 N m

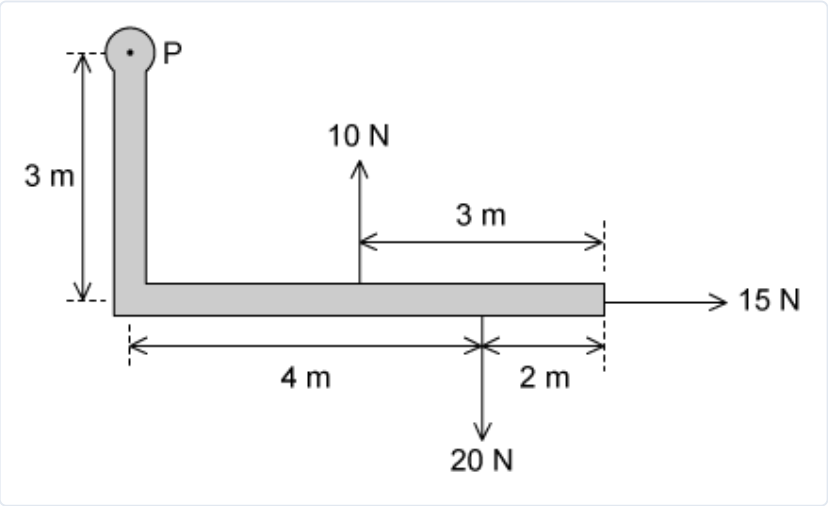
B 0.98 N m

C 6.1 N m

D 9.6 N m

Question 14 Medium [1 mark]

A rigid L-shaped lever arm is pivoted at point P.



Three forces act on the lever arm, as shown in the diagram.

What is the magnitude of the resultant moment of these forces about point P?

A 5 N m

B 65 N m

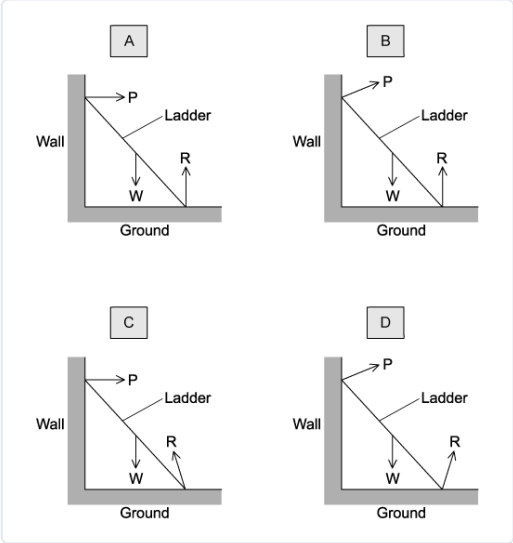
C 95 N m

D 155 N m

Question 15 Medium [1 mark]

A ladder is positioned on icy ground and is leant against a rough wall. At the instant of release, it begins to slide.

Which diagram correctly shows the directions of the forces *P*, *W* and *R* acting on the ladder as it slides?



A

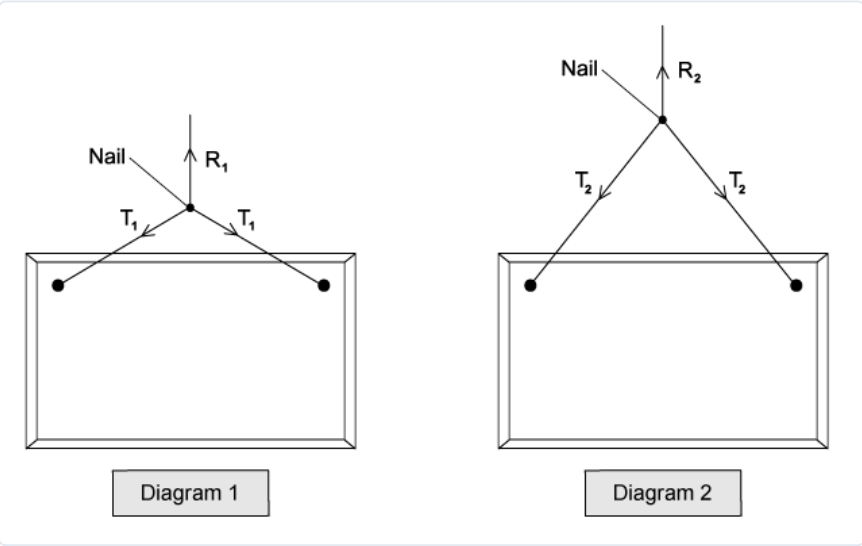
B

C

D

Question 16 Hard [1 mark]

The diagrams show two ways of hanging the same picture.



In both cases, a string is attached to the same points on the picture and looped symmetrically over a nail in a wall. The forces shown are those that act on the nail.

In diagram 1, the string loop is shorter than in diagram 2.

Which information about the magnitude of the forces is correct?

A	$R_1 = R_2$	$T_1 < T_2$
B	$R_1 = R_2$	$T_1 > T_2$
C	$R_1 > R_2$	$T_1 < T_2$
D	$R_1 < R_2$	$T_1 = T_2$

A

B

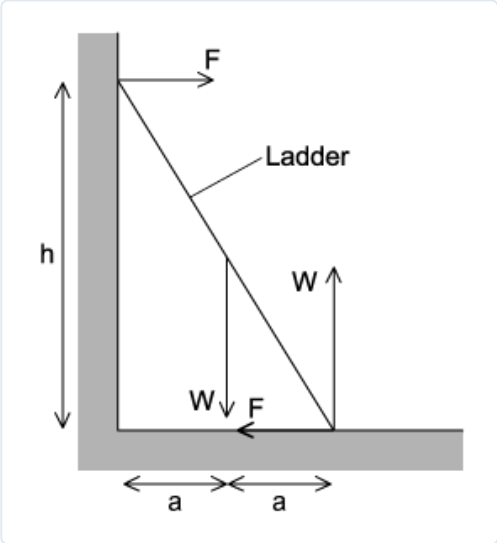
C

D

Question 17 Hard [1 mark]

A uniform ladder rests against a smooth vertical wall, as shown in the diagram. The bottom of the ladder rests on rough ground where friction is not negligible. The top of the ladder is a vertical height h up the wall, and the foot of the ladder is a horizontal distance $2a$ from the wall.

The weight W of the ladder acts at its centre. At the foot of the ladder, the ground exerts a vertical reaction R and a horizontal frictional force F.



Which equation represents the moments when the ladder is in equilibrium?

A $Wa + Fh = 2Wa$

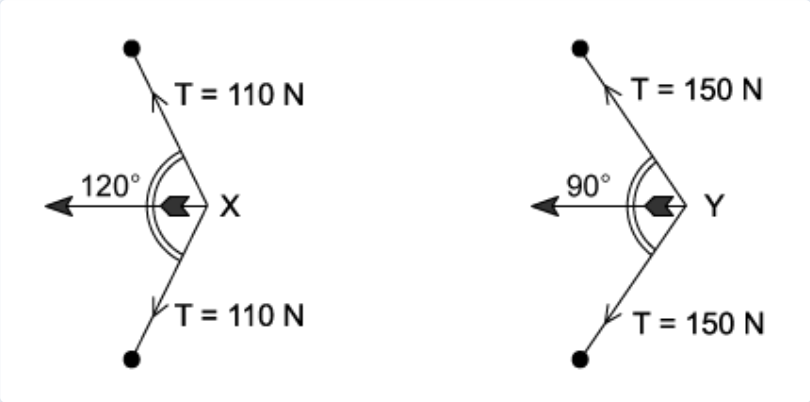
B $Fa + Wa = Fh$

C $Wa + 2Wa = Fh$

D $Wa - 2Wa = 2Fh$

Question 18 Hard [1 mark]

An archer draws his bowstring back to position X. The bowstring and arrow are shown. The tension *T* in the string is also shown. Then he draws the bowstring back further to position Y.



The archer finds the resultant force on the arrow is greater when the arrow is released from position Y.

What is the increase in force?

A 51 N

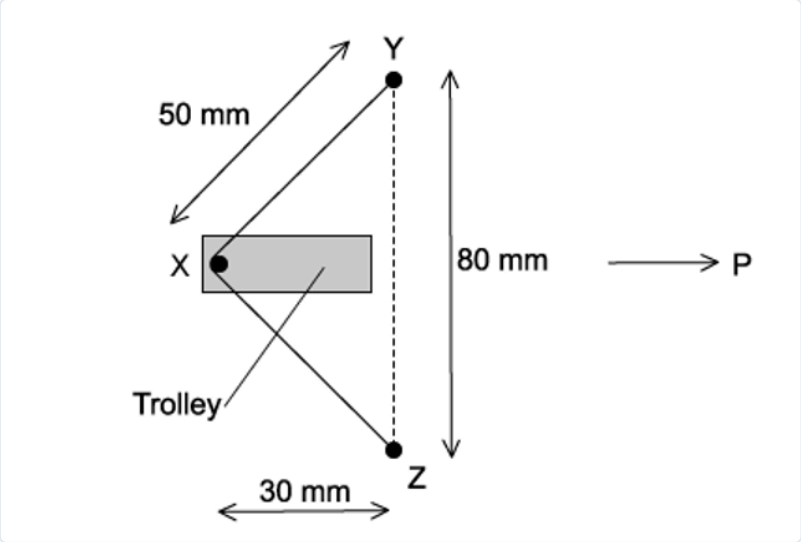
B 55 N

C 102 N

D 110 N

Question 19 Hard [1 mark]

The diagram shows two fixed pins, Y and Z. A length of elastic is stretched between Y and Z and around pin X, which is attached to a trolley.



X is at the centre of the elastic and the trolley is to be propelled in the direction P at right angles to YZ. The tension in the elastic is 4 N.

What is the force accelerating the trolley in the direction P when the trolley is released?

A 2.4 N

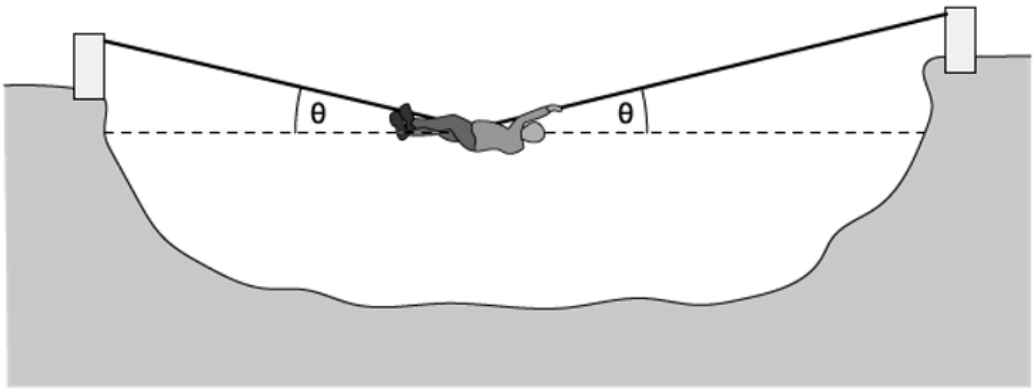
B 3.2 N

C 4.8 N

D 6.4 N

Question 20 Hard [1 mark]

The diagram shows a rope bridge that a student makes on an adventure training course. The student has a weight *W*.



Which formula gives the tension *T* in the rope?

A $T = \frac{W}{2\cos\theta}$

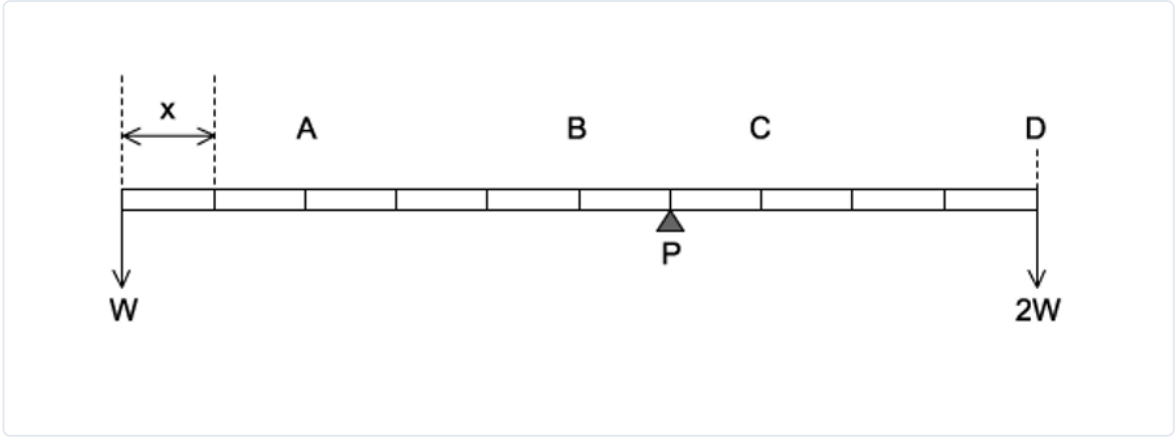
B $T = \frac{W}{2\sin\theta}$

C $T = \frac{W}{\cos\theta}$

D $T = \frac{W}{\sin\theta}$

Question 21 Hard [1 mark]

A uniform beam is pivoted at P as shown. Weights of *W* and 2 *W* are attached to its ends. The beam is marked at *x* intervals. The weight of the beam is 10 *W*



At which point should a 2 *W* weight be attached to achieve equilibrium?

A

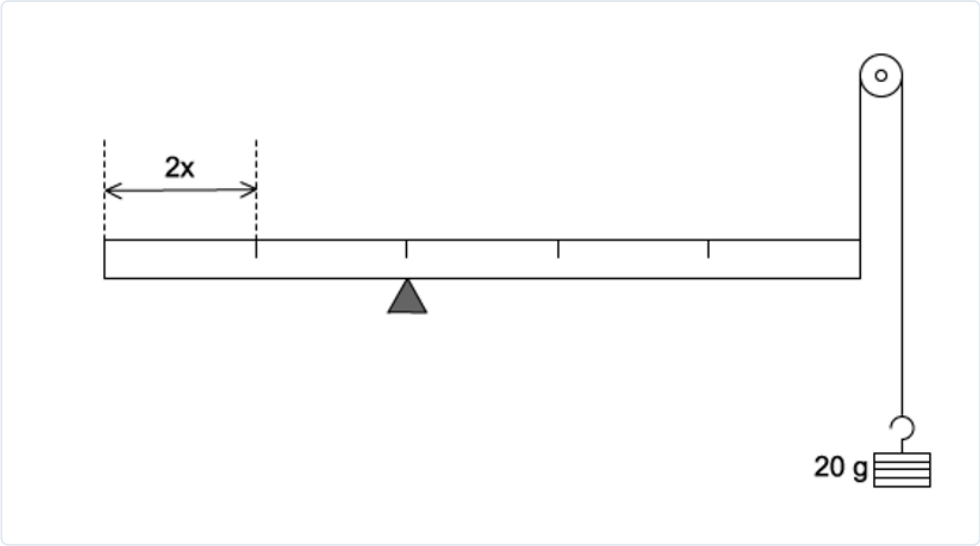
B

C

D

Question 22 Hard [1 mark]

A uniform metre rule of mass 100 g is supported by a pivot and a string as shown in the diagram. The beam is marked at intervals of $2x$. The string passes round a frictionless pulley and carries a mass of 20 g.



At which mark on the rule must a 50 g mass be suspended so that the rule balances?

A $0.4x$

B $3.6x$

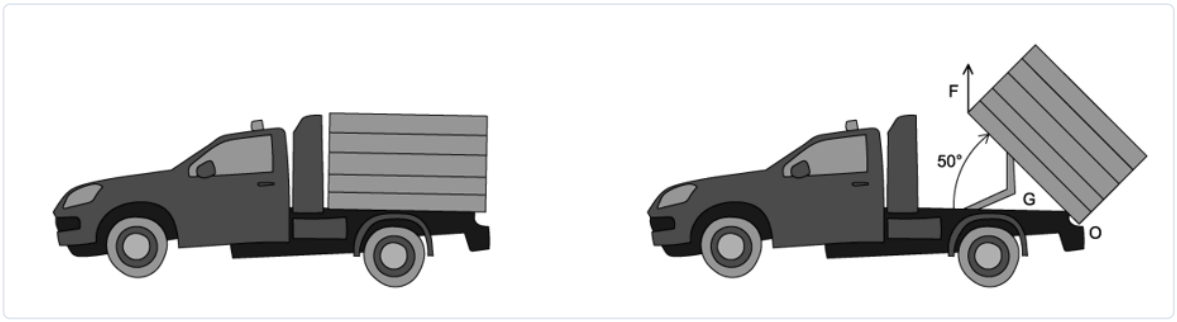
C $4.4x$

D $6.4x$

Question 23 Hard [1 mark]

A trailer of weight 30 kN is attached to a truck. The trailer is raised until it is held at 50˚ to the horizontal by a vertical force *F* as shown in the diagram.

The trailer is 2.7 m in length and the position of its centre of gravity G is a third of the length from the hinge at O.



What is the vertical force *F* applied to the end of the trailer?

A 10 kN

B 15 kN

C 30 kN

D 60 kN